WORKED SOLUTION

STEP III 2007, Question 6

Pure • Complex Numbers and Circles

STEP III 2007, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q6

The key idea

On a circle of radius a centred at the origin, conjugation is the same as inversion: $pp^* = a^2$, so $p^* = a^2/p$. That one fact turns every conjugate in the problem into an ordinary fraction, and the identity drops out. For perpendicularity, the test is that the ratio of the two chord vectors is purely imaginary. The identity then turns this test into a simple product condition. For the polygon, that product condition is a recurrence: once B_1 is chosen, every other B_k is forced, and the only question is whether the chain closes up when it gets back round to B_1.

Solution

The identity

Since P lies on the circle, $|p| = a$, so $pp^* = a^2$ and $p^* = a^2/p$; similarly $q^* = a^2/q$. Hence

$$p^* - q^* = \frac{a^2}{p} - \frac{a^2}{q} = \frac{a^2(q-p)}{pq} = -\frac{a^2(p-q)}{pq}.$$

This is nonzero because $p \neq q$ (and $p, q \neq 0$ since $a > 0$), so we may rearrange:

$$pq = -a^2 \frac{p-q}{p^* - q^*},$$

as required.

Perpendicular chords

The chord PQ is in the direction of the complex number $p - q$, and RS in the direction of $r - s$. Two nonzero complex numbers are perpendicular exactly when their ratio

$$w = \frac{p-q}{r-s}$$

is purely imaginary, because the argument of the ratio is the angle between them.

Ask yourself How do we turn "w is purely imaginary" into an equation we can use?

A nonzero w is purely imaginary if and only if $w^* = -w$, that is, $w/w^* = -1$. Using the identity for both chords (the second with r, s in place of p, q),

$$\frac{w}{w^*} = \frac{p-q}{p^*-q^*} \cdot \frac{r^*-s^*}{r-s} = \frac{-pq/a^2}{-rs/a^2} = \frac{pq}{rs}.$$

So $PQ \perp RS$ if and only if $pq/rs = -1$. Since $rs \neq 0$, that means

$$pq + rs = 0.$$

Every step here was reversible, so we have shown slightly more than was asked: for chords of this circle, $PQ \perp RS \iff pq + rs = 0$. We need the converse direction below.

The polygon problem

Let the points A_k and B_k be represented by α_k and β_k. By the previous part (and its converse), the n perpendicularity conditions are exactly

$$\beta_k\beta_{k+1} = -\alpha_k\alpha_{k+1} \quad (k = 1, \ldots, n-1), \qquad \beta_n\beta_1 = -\alpha_n\alpha_1,$$

provided consecutive Bs are distinct, so that each B_kB_{k+1} really is a chord.

Choose β_1. The first $n-1$ conditions then force $\beta_2, \beta_3, \ldots, \beta_n$ in turn, since $\beta_{k+1} = -\alpha_k\alpha_{k+1}/\beta_k$, and each β_{k+1} automatically has modulus a. Working out the first few:

$$\beta_2 = -\frac{\alpha_1\alpha_2}{\beta_1}, \qquad \beta_3 = -\frac{\alpha_2\alpha_3}{\beta_2} = \frac{\alpha_3\beta_1}{\alpha_1}, \qquad \beta_4 = -\frac{\alpha_3\alpha_4}{\beta_3} = -\frac{\alpha_1\alpha_4}{\beta_1},$$

and an easy induction gives

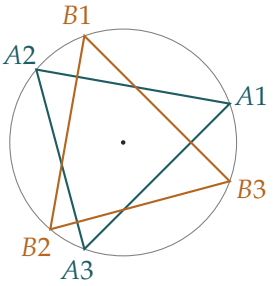
$$\beta_k = \frac{\alpha_k\beta_1}{\alpha_1} \ (k \text{ odd}), \qquad \beta_k = -\frac{\alpha_1\alpha_k}{\beta_1} \ (k \text{ even}).$$

(If it holds for k, then $\beta_{k+1} = -\alpha_k\alpha_{k+1}/\beta_k$ gives the other form for $k+1$.) Everything now depends on whether the last condition, $\beta_n\beta_1 = -\alpha_n\alpha_1$, holds.

$n = 3$. Here $\beta_3 = \alpha_3\beta_1/\alpha_1$, so the closing condition is

$$\frac{\alpha_3\beta_1^2}{\alpha_1} = -\alpha_3\alpha_1 \iff \beta_1^2 = -\alpha_1^2 \iff \beta_1 = \pm \mathrm{i}\,\alpha_1.$$

So there are only two choices of B_1. With $\beta_1 = \mathrm{i}\alpha_1$ the formulae give $\beta_2 = \mathrm{i}\alpha_2$ and $\beta_3 = \mathrm{i}\alpha_3$; with $\beta_1 = -\mathrm{i}\alpha_1$, every $\beta_k = -\mathrm{i}\alpha_k$. In each case the triangle $B_1B_2B_3$ is the triangle $A_1A_2A_3$ rotated through a right angle about the centre. Its vertices are distinct, so both choices genuinely work, and rotating a chord through 90° certainly makes it perpendicular to the original.



The figure shows the choice $\beta_1 = \mathrm{i}\alpha_1$: each side of the amber triangle is the corresponding teal side turned through $90°$.

$n = 4$. Now $\beta_4 = -\alpha_1\alpha_4/\beta_1$, so

$$\beta_4\beta_1 = -\alpha_1\alpha_4$$

holds automatically, whatever β_1 is. So B_1 can be chosen *anywhere* on the circle, giving infinitely many choices. The only exceptions are the finitely many positions in which two consecutive Bs would coincide. For instance, $\beta_1 = \beta_2$ needs $\beta_1^2 = -\alpha_1\alpha_2$, which rules out two points. Each of the four coincidences rules out at most two points, so at most eight positions are excluded.

$n > 4$. The same calculation works for every n, and only the parity of n matters:

- if n is odd, the closing condition is $\beta_1^2 = -\alpha_1^2$. There are exactly two choices, $\beta_1 = \pm\mathrm{i}\alpha_1$, and $B_1B_2\cdots B_n$ is the polygon $A_1A_2\cdots A_n$ rotated through $\pm 90°$ about the centre;
- if n is even, the closing condition holds automatically. B_1 may be any point of the circle apart from at most $2n$ positions that would make consecutive Bs coincide, so there are infinitely many choices.

Checks

- Try $p = a$, $q = \mathrm{i}a$: then $pq = \mathrm{i}a^2$, and $-a^2(a - \mathrm{i}a)/(a + \mathrm{i}a) = -a^2 \cdot (-\mathrm{i}) = \mathrm{i}a^2$, as it should be.
- Try the diameters $p = a$, $q = -a$ and $r = \mathrm{i}a$, $s = -\mathrm{i}a$, which are perpendicular: $pq + rs = -a^2 + a^2 = 0$.
- Numerically, we tested the identity and the criterion $PQ \perp RS \iff pq + rs = 0$ on 1000 random configurations. For $n = 3, \ldots, 9$ we ran the recurrence from random starting points. For odd n, scanning B_1 round the circle in $0.1°$ steps, the chain closed only near $\beta_1 = \pm\mathrm{i}\alpha_1$. For even n it closed every time, with all chords perpendicular to rounding error.

Common slips

- Proving only "perpendicular $\Rightarrow pq + rs = 0$". For the polygon we need the converse as well, to know that the βs we construct really do give perpendicular chords.
- Writing "perpendicular means w is imaginary" as $w + w^* = 0$ without noting $w \neq 0$. Here $w \neq 0$ is guaranteed because the points are distinct.
- For even n, saying "every B_1 works" without noticing that a few positions make two consecutive Bs coincide, so that the "chord" has no direction.

Beyond STEP

The map $z \mapsto a^2/z^*$ is *inversion* in the circle $|z| = a$, and the fact that points on the circle are fixed by it ($p = a^2/p^*$) is what powered this whole question. Inversion is one of the Möbius transformations, which are central to a first course in complex analysis and geometry. The polygon argument has the flavour of *monodromy*: follow a chain of forced choices round a loop and ask whether you return to where you started, and the answer depends on the parity of n. Needham's *Visual Complex Analysis* develops inversion and these geometric ideas beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.